

AI for Economic Research: Dynamic Models, Language, and Agents

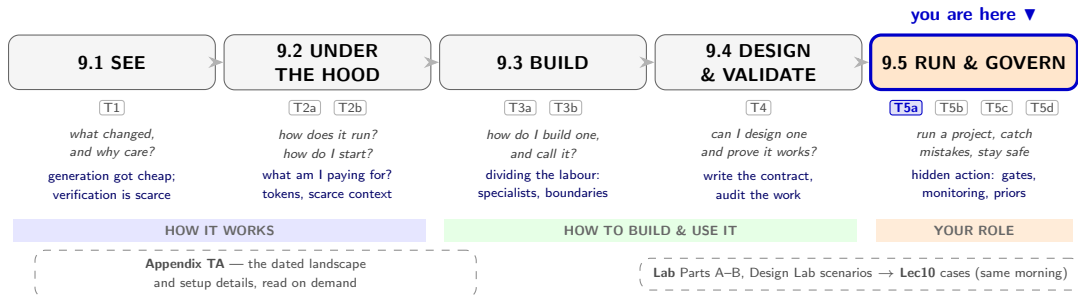
Lecture 9.5a: Method — The Homotopy Workflow

Zhigang Feng

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Lecture 9 in One Map: The Homotopy Method

This deck opens the last part: how do I structure real work with the agents I just learned to build?



Route: the homotopy idea → the six-step pipeline, step by step (with HA-Ramsey's actual opening prompts) → how to grade a plan before approving it → the runnable OLG demo → which tool for which step. The economist's question for Part 9.5: how do you manage hidden action over months — gates, monitoring, and your own prior?

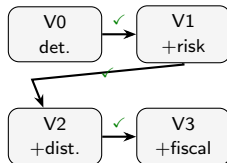
The Homotopy Workflow

Build from simple to complex, validate, then extend

The Homotopy Idea: Solve the Easy Version First

In workflow design, homotopy means solving an easier version first and then extending it toward the full research task.

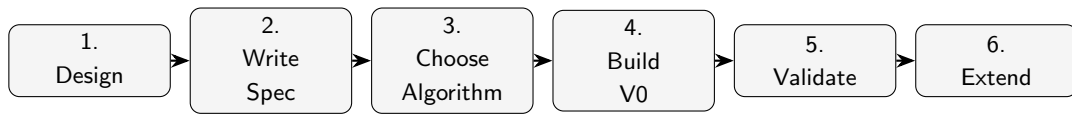
- Start from a version with clear intuition and known checks
- Use that validated baseline as the starting point for the next
- Attribute new bugs to the new margin you just added
- Never ask the agent to solve the full complicated problem in one jump



Each step small; each step validated; if a step fails, return to the last validated version.

Why the name: homotopy continuation starts from a problem with a known solution and *continuously deforms* it toward the target (Judd 1998, Ch. 5–7). $V0 \rightarrow V1 \rightarrow V2$ are gates along that deformation — the same discipline, scaled up, produced HA-Ramsey (Story 1; case in Lec10 T8).

Homotopy in Practice: Build from Simple to Complex



This replaces the fantasy prompt “solve my project.”

- Start with a problem you can state and benchmark
- Give the agent bounded tasks with visible acceptance criteria
- Add one new ingredient at a time

The next frames walk the six steps in order — with the HA-Ramsey case showing what each step looks like at research scale.

Step 1 — Design: State the Goal Like an Economist

Before any code, write down the task in economist language.

- Research question in one sentence
- Objective and constraints, or the relevant source rules
- Equilibrium definition, estimand, or review standard if applicable
- Parameters, sample, source universe, and economic motivation
- Outputs you will treat as evidence

Aiyagari anchor:

$$\max \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t) \quad \text{s.t.} \quad c_t + a_{t+1} = (1+r)a_t + wz_t, \quad a_{t+1} \geq \underline{a}$$

Goal template (T2b's pen-and-paper step, made precise): *"Compute X with method Y to answer Z; success looks like W."* If you cannot fill in W, you are not ready to open the terminal.

The 5-Pillar Spine: What You Supply Before Step 2

The five pillars (introduced in Lec01, exercised throughout Lec06) reappear here as the agent-era research workflow.

1. **Theory**
2. **Algorithm**
3. **Numerics**
4. **Advanced**
5. **Code literacy**

The agent accelerates each pillar; it replaces none of them.

Rendering	One-line discipline
1. Question design	Economic language; define the evidence; limiting cases first
2. Procedure design	The right procedure <i>and</i> its acceptance criterion, before code
3. Data/numerical discipline	Grids, weights, crosswalks, missingness — saved, not implicit
4. Domain constraints	Know when the agent's convenient default is wrong
5. Code literacy	Read what was built; verify it is what you meant

The throughline: pillars 1–4 decide *what should be built and what counts as success*; the machine only decides how fast. Weak pillars = faster nonsense.

Step 2 — The Spec Is the Contract

The specification file is the contract between you and the agent.

A good specification file states four things clearly:

1. assumptions and domain rules
2. data or source restrictions
3. procedure and validation target
4. required outputs and where to save them

Examples: CRRA utility and convergence tolerance for Aiyagari; weights and subgroup cuts for MEPS; quote-level evidence rules for paper review; official/institutional-source rules and provenance fields for fellows data.

Best practice: ask the agent to restate the specification in its own words before it writes code. That catches misunderstandings when they are still cheap.

Effective Prompting: Vague vs. Specific

A specific prompt names the object, the method, and the check that tells you it worked. A vague one leaves all three to the agent.

Vague	Specific	The acceptance check
"Analyze this data."	Load <code>employment.csv</code> ; compute monthly growth by sector; plot a time series with a source note	growth rates match a hand-computed sector; source and vintage printed
"Make it faster."	Vectorize the inner VFI loop with NumPy broadcasting; profile before and after	same policy function to 10^{-10} ; the profile shows the speedup
"Fix the regression."	Cluster standard errors at the level where the treatment is assigned	the clustering variable is named in the code and the table note
"Add some validation."	Add an Euler-residual check; report the max and the mean	both below 10^{-6} on the grid, printed every run

The third column is the one agents cannot supply for you. A specific prompt without an acceptance check is still a request for something that merely looks finished.

How HA-Ramsey Opened: Restate, Audit, Refine

The restate-first practice is not hypothetical — these three prompts opened the HA-Ramsey collaboration.

1. "Here are the RA paper (TeX) and the RA repository.
Restate the model and the algorithm in your own words.
List every equation and where it is implemented."
2. "Audit the code against the document. Report every place
where the code and the TeX disagree: scoring, bounds,
sampling, penalties. Classify each as a correctness risk
or a documentation gap."
3. "Propose a refined RA document + code pair in which the
two are consistent. Do not add any new economics yet."

- Prompt 1 tests comprehension; prompt 2 tests the *seed*; prompt 3 fixes the foundation while freezing the economics
- “The audit is the cheapest insurance in the whole project” — extending an inconsistent seed propagates the inconsistency

Grade the Plan Before Any Code: Five Plan-Grading Criteria

T2b said: start in plan mode, review, approve. Approve against what? These five criteria.

1. **Restates your goal correctly** — in its own words, not your words echoed back
2. **Bounded steps** — each step has a named deliverable small enough to review in minutes
3. **Acceptance criterion per step** — a benchmark, test, or check named *in advance*
4. **Stop and rollback points** — what happens when a step fails; which validated version you return to
5. **Smallest testable object first** — the plan begins with V0, not with the full target

A plan without acceptance criteria is a wish. Send it back — a plan revision costs one message; a wrong build costs a day.

Fellows instance: one batch of 50 = a bounded step; “every filled field carries a source URL” = its acceptance criterion.

HA-Ramsey instance: the audit prompt is criterion 3 applied to the seed itself.

Step 3 — Algorithm and Pseudocode Catch Design Errors

Pseudocode or a phase plan is where you catch design mistakes before implementation.

```
Phase A: ingest or define the object  
Phase B: run the chosen procedure  
Phase C: save diagnostics and checks  
Phase D: report only what passed validation
```

Concrete procedures: VFI or EGM for Aiyagari, a 3-prompt pipeline for MEPS, section decomposition for paper review, and batched metadata collection for fellows data.

The point is not elegance. The point is that you and the agent agree on the procedure before code details begin to hide conceptual errors.

Step 4 — V0: The Simplest Version With a Benchmark You Understand

V0 is the simplest version with a benchmark you understand — ideally a closed form.

Optimal growth V0: full depreciation. With log utility, Cobb–Douglas output $y_t = z_t k_t^\alpha$, and $\delta = 1$, the model has an exact closed-form solution:

$$k_{t+1} = \alpha\beta z_t k_t^\alpha, \quad c_t = (1 - \alpha\beta) z_t k_t^\alpha$$

- Your V0 solver must reproduce this policy to numerical precision — an *absolute* benchmark, not a plausibility check
- Then relax $\delta < 1$: the closed form disappears, but the validated V0 has already certified the plumbing — grids, interpolation, Euler equation, simulation

Discipline: do not ask the agent to add partial depreciation, risk, or equilibrium search before V0 matches the closed form.

V0 at Research Scale: What Makes a Good Seed

At research scale, V0 has a name: the seed. HA-Ramsey grew from a published RA solver because that seed had four properties.

1. **It works** — reproduces its own paper's numbers, not just “runs without error”
2. **It is documented** — a TeX model note in the *same notation* as the code
3. **It is modular** — config-driven, so every later experiment is a config diff, not a code fork
4. **It points somewhere** — the RA model is the degenerate no-risk case of the HA target; the deformation path exists by construction

A good seed is like a good prototype for manufacturing a new product: it already works end-to-end at small scale, so scaling up is engineering and refinement — not a fresh invention on the factory floor.

Same object, different scales: the classroom V0 and the research seed play the same role. The OLG demo's v0 (later this deck) is your training-wheels seed.

What “V0 First” Means Across Four Tasks

Task	V0	First validation target
Aiyagari model	deterministic household problem	recover simple savings logic and boundary behavior
MEPS pipeline	3-year pilot panel	weighted rates line up with an external benchmark
Paper review skill	one section or theorem block	quotes are real and comments are specific
Fellows dataset	seed roster only	source rules and provenance fields are actually recorded

Why economists should care: V0 keeps the first error interpretable. You want the first failure to be obvious, not hidden inside a giant workflow.

Steps 5–6 — Validate, Then Extend One Margin at a Time

Version	New ingredient	Required benchmark
V0	Deterministic household	analytical cases
V1	One new model, data, or review margin	setting the new margin to zero recovers V0
V2	Saved diagnostics and state tracking	mass sums, benchmarks, quote checks, or source logs behave sensibly
V3	Full target workflow	final artifact survives validation and interpretation

Why this works with agents: when a new version breaks, both you and the agent know where to look.

Hold the gate idea: in T5b it returns as a management principle — a quality gate on every intermediate good, not just on code versions.

Validation Is Your Referee Checklist

Every iteration needs both workflow checks and economic checks.

Workflow checks

- Convergence achieved
- No NaN or Inf values
- Mappings, weights, or source logs saved where expected
- Diagnostics, residuals, or quote checks pass
- Script reproduces the same output twice

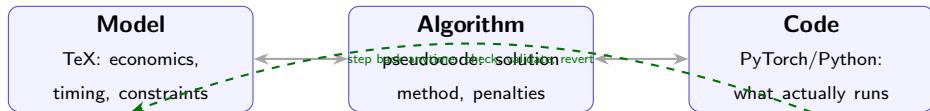
Economic checks

- Limiting cases recover known results
- Comparative statics, subgroup patterns, or review claims make sense
- Magnitudes and source coverage are plausible
- Inference uses the right standard errors, sample, and restrictions
- You can explain the result in plain economic language

If you cannot defend the output in a seminar, you are not done.

Three Layers Grow Together: Model, Algorithm, Code

A research project is not one artifact but three, and the homotopy steps move all three in lockstep.



- **Forward:** you grow from the current version toward the final version gradually — a new model element forces a new algorithm forces new code, one margin at a time
- **Backward:** every earlier version is preserved and validated, so you can always step back to a simpler one — to check and validate a result, to isolate where something broke, or to revert to the last working version
- **The agent's job here:** keep TeX, pseudocode, and code describing *the same economy* at every version — drift between layers is where silent errors live

Why This Beats One-Shot Generation

Four reasons the homotopy workflow is especially effective with agents:

1. **Agents excel at well-specified, bounded tasks.**

“Solve my model” fails. “Implement V0 per this pseudocode and validate against these 3 benchmarks” succeeds.

2. **Starting simple gives both you and the agent a working codebase.**

V1 inherits validated V0 code. The agent builds on tested functions, not a blank file.

3. **Bug attribution is immediate.**

If V2 breaks, the bug is in V2's new lines, not the validated inherited ones.

4. **Context builds naturally.**

By V3, the agent has seen the spec, pseudocode, three validated versions, and every correction.
Deep context — not a 10-page prompt.

Discipline: resist the urge to skip ahead. A working V0 takes 30 minutes; debugging a broken V2 because V0 was never validated takes a day. This is also how good RAs are trained: simple task, verify, widen scope.

OLG Five-Step Demo

A concrete homotopy workflow for computational macro

Demo: Folder Structure Is the Method

Folder or file	Teaching role
stages/	five durable stages: model, equilibrium, algorithm, pseudocode, implementation/validation
prompts/v*_to_v*.md	one bounded terminal-agent session per version transition
versions/v0-v5	validated working code preserved at every step
tests/	one smoke test per version; failures identify the new margin
build/*.json	reproducibility reports from terminal runs

Demo path: `demos/M4_olg_5step_demo/` (local copy in this lecture; mirror of `MiniCourse_8hr M4_olg_5step_demo` minus `.venv/build/`).

Note the pattern: the `HA-Ramsey Case_Study/` package (Lec10 T8) is this same folder discipline at paper scale — seed, versions, notes, audits, all preserved.

Version Ladder: One Economic Margin at a Time

Ver.	Adds	Validation gate
V0	3 cohorts, 1 asset, 2-state TFP, one NN policy	deterministic steady-state collapse; procyclical capital; RMS Euler residual below threshold
V1	7 cohorts and lifecycle labor	V0 logic preserved; middle-aged consumption peak appears
V2	4-state Markov TFP	collapse back to two states reproduces V1
V3	bonds, price, clearing, complementarity	zero bond-loss weight reproduces V2; positive late-life holdings
V4	capital adjustment cost	zero adjustment cost reproduces V3; capital path smooths
V5	stabilizing homotopy phases	residuals fall by phase; final Euler error target met

Research lesson: the agent never receives “solve the big model.” It receives one validated deformation at a time.

Demo Launch and Classroom Contract

```
cd source/Lec09_Agentic_AI/demos/M4_olg_5step_demo
pip install -e .
jupyter notebook demo.ipynb          # Run All: V0 -> V5 with each gate
python3 -m unittest discover -s tests # separate smoke check
```

- **What the demo proves:** homotopy preserves a working baseline while complexity rises.
- **What to inspect:** prompts, version folders, test names, validation messages, and regenerated reports.
- **Failure recovery:** if V3 breaks, return to validated V2 and add only the bond-market-clearing layer.

Tool Choice

Match the tool to the pipeline step

Tool Choice per Pipeline Step

The six-step pipeline also tells you which tool to open.

Tool type	Pipeline steps	Economic example
Web chat	1–2: design, draft the spec	clarify identification logic, draft a model summary
IDE assistant	single-file edits inside a step	vectorize one function, explain one traceback
Terminal agent (Claude Code, Codex CLI, Gemini CLI)	4–6: build, validate, extend	restructure the project, run scripts, generate diagnostics, iterate

Why the terminal agent becomes the workhorse: it runs the real code on the real project, while leaving you in the approval and validation loop — enough autonomy to save time, not so much that you stop checking the economics.

The OLG demo ships both `CLAUDE.md` and `AGENTS.md`: one project, briefed for more than one harness.

Bridge: One Ladder vs. the Whole Factory

T5a disciplined the growth of one artifact. A real project is a factory: many artifacts, many sessions, and a team of the agents you built in Part 9.3.

- You now have the *method*: design, spec, algorithm, V0, validate, extend — with a plan graded before any code
- But a paper is model + algorithm + code + data + reviews + write-up, worked across weeks of sessions
- The next question: “*how do I run the whole project?*” — T5b’s answer is a role change: the researcher as **project manager**, and the project files (instructions, skills, agents, rules) as the manager’s instruments

Arc: 9.1 See → 9.2 Under the Hood → 9.3 Build → 9.4 Design & Validate → **9.5 Run & Govern.**

Appendix: A Concrete Aiyagari Spec and Pseudocode

Steps 1–2 for a model this course already uses: the spec the agent reads every session, and pseudocode agreed before any Python.

`model_spec.md` — a minimal version

1. **Household:** CRRA utility, $\beta = 0.96$, $\sigma = 2$
2. **Income:** AR(1) in logs, $\rho = 0.9$, $\sigma_\varepsilon = 0.2$, Tauchen with 7 states
3. **Asset grid:** 200 points, spaced densely near the borrowing limit
4. **Method:** VFI with Howard improvement
5. **Convergence:** sup-norm on V below 10^{-8}
6. **Equilibrium:** r clears the asset market, by bisection
7. **Output:** $c(a, z)$, $a'(a, z)$, the stationary distribution, the Gini coefficient

Then ask the agent to restate the model in its own words.

Pseudocode for V0 (deterministic income)

```
grid a[1..N]; guess V_old = u(r*a + w) / (1 - beta)
repeat until ||V_new - V_old|| < tol:
  for each a_i:
    for each a' with c = (1+r)*a_i + w - a' > 0:
      value = u(c) + beta * interp(V_old, a')
      V_new[i] = max over a'
      a_policy[i] = argmax over a'
  every 25 iterations: 50 Howard steps
return V_new, a_policy, c = (1+r)*a + w - a_policy
```

Note the $c > 0$ guard and the stopping rule — the two things the lab's flawed solver leaves out (T4, scenario S1).

Appendix: Three Analytical V0 Benchmarks

Before V1, the deterministic-income V0 must reproduce three cases you can derive on paper. If it fails one, nothing built on it can be trusted.

Calibration	What theory says	What V0 must show
$\beta(1+r) = 1$	the Euler equation holds with constant consumption	$a'(a) = a$: the policy lies on the 45-degree line
$\beta(1+r) < 1$	impatience outweighs the return: dissave	assets run down to the borrowing limit
$\beta(1+r) > 1$	the return outweighs impatience: save	assets keep growing toward the top of the grid

If V0 fails: the Bellman equation, the grid, or the stopping rule is wrong — and every later version inherits it. A working V0 takes an afternoon; debugging a V2 built on a broken V0 takes a week.